Multi-User Collaborative Air Drawing on a Single Webcam

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1. Problem Description

Collaborative whiteboarding traditionally requires either a single physical board — where participants take turns — or large multi-touch displays that are prohibitively expensive for small classrooms and home settings. Webcam-based air-drawing is a low-cost alternative, but existing systems focus almost exclusively on **single-user** interaction. When two or more users share one camera, three failure modes emerge: (i) hands **occlude one another** when reaching across the frame; (ii) hands **cross paths** while drawing, causing trackers to swap identities; and (iii) users **exit and re-enter the frame** between strokes, after which the system can no longer tell who is who.

We build a real-time system that lets two to three users draw simultaneously on a shared canvas using one commodity 720p webcam, with every stroke correctly attributed even under occlusion, crossing, and re-entry. The core problem is **robust per-user hand tracking and re-identification on a monocular video stream** — a multi-object tracking (MOT) problem specialized to the hand domain, where appearance cues are weaker than in person-tracking and motion is highly non-rigid.

2. Method

(a) Hand detection. MediaPipe Hands [Zhang et al., 2020] returns 21 landmarks per hand at 30 FPS on CPU. Because its default configuration is limited to two simultaneous hands, we fine-tune **YOLOv8-hand** on EgoHands and 11k Hands (plus our own captures) to detect up to six hands per frame; landmark regression then runs on each crop.

(b) Multi-hand tracking with re-identification. Bounding boxes feed **ByteTrack** [Zhang et al., ECCV 2022], which performs two-stage association via Kalman-filtered IoU to handle partial occlusion. Because hands lack the distinctive appearance of full bodies, motion + IoU alone cannot recover identity after long occlusion. We attach an **appearance head**: each hand crop is encoded by a MobileNetV3 backbone into a 128-D embedding, trained with triplet loss on a self-collected set (~10 subjects, varied lighting). Lost tracks are re-linked to new detections by cosine similarity above threshold.

(c) User binding. Each user performs a calibration wave at session start, binding the tracker ID to a user ID, color, and pen profile. A state machine handles users joining mid-session via the same gesture.

(d) Drawing pipeline. The index fingertip (landmark 8) defines each user's cursor; a thumb–index pinch — distance between landmarks 4 and 8 below an adaptive threshold normalized to hand size — toggles pen-down/up, resolving monocular depth ambiguity. A 1€ filter smooths fingertip jitter without visible latency. Strokes render on the shared canvas in the owner's color and are logged with millisecond timestamps for Z-ordering and conflict analysis.

Evaluation protocol. We record ~10 multi-user sessions (≈30 min) covering staged scenarios — side-by-side drawing, hand-crossing, arm-occlusion, and frame re-entry — and annotate hand bounding boxes and user IDs using CVAT. We ablate three configurations — (i) IoU-only, (ii) ByteTrack, (iii) ByteTrack + appearance re-ID — and report MOT metrics (MOTA, IDF1) alongside our task-specific metric.

3. Expected Result

A working real-time demo supporting up to three simultaneous users on a single 720p webcam, sustaining **≥25 FPS** on a consumer laptop (no discrete GPU required). The primary task-specific metric is **stroke-attribution accuracy** — the fraction of drawn strokes correctly assigned to the user who produced them. We target:

- **≥85% stroke-attribution accuracy** under cross-and-occlusion scenarios;
- **IDF1 ≥ 0.80** for hand identity preservation across the full session;
- The appearance-augmented ByteTrack configuration outperforming IoU-only by **≥15 percentage points**, demonstrating that the appearance head is the load-bearing component for the multi-user setting.

Deliverables: (1) the real-time system with a simple GUI showing the shared canvas and per-user color legend; (2) the annotated multi-user evaluation dataset; (3) an ablation report across MOT and task-specific metrics with qualitative failure-case analysis. A stretch goal is a small user study (3–5 groups) measuring perceived attribution correctness and collaborative satisfaction.

References

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